From Theory to Deep Space: Real-Data Validation of the AIKC Extension for IBI via NASA ACE Mission

A New Paradigm in Information-Based Metrology: Plasma-Resilient Control for Spacecraft and CubeSats

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Abstract

The effective theory of Information-Based Instrumentation (IBI) established a rigorous framework for precision metrology on edge devices, integrating Mori–Zwanzig projections, Fisher geometry, and 3-state Kalman filtering. However, its domain of applicability is explicitly limited to linear regimes and Gaussian fluctuations. This work presents the experimental implementation and validation of Adaptive Information Kernel Control (AIKC), a modular layer that extends the operational envelope of IBI to nonlinear media with heavy tails and intermittent shocks, without altering its theoretical foundations. Through exhaustive processing of 210,216 high-resolution magnetometry measurements (Bz component, MFI Level-2 product, year 2023) at 150 s target cadence from the NASA ACE mission, it is demonstrated that AIKC, with the Fisher kernel as its exponent, achieves superior accuracy (MSE ≈ 0.0415, NSE ≈ 0.9964, MBE $\approx +1.1\times10^{-5}$) compared to the Adaptive Kalman Filter (AKF) (MSE ≈ 0.6969, NSE ≈ 0.9390, MBE $\approx +1.94\times10^{-3}$) and offers significantly better precision relative to PINN (MSE ≈ 0.0942, NSE ≈ 0.9918, MBE $\approx$-2.76×10^{-4}) while maintaining a $\sim$60% reduction in total computational load compared to the neural alternative. The information kernels exhibit contractive stability, negligible systematic bias ($|\text{MBE}| < 2\times10^{-4}$), and preservation of the Cramér–Rao bound under high-frequency space plasma conditions. These results validate the IBI-AIKC framework as a viable architecture for high-reliability space missions and CubeSats.

Keywords: Information-Based Instrumentation (IBI), Adaptive Information Kernel Control (AIKC), Experimental Validation, NASA ACE Mission, Space Plasma, Fisher Geometry, Computational Efficiency, Information Kernels, Kalman Filtering, PINNs, Adaptive Hysteretic Matter (AHM), Spacecraft, CubeSats.

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1 Introduction

Deep space exploration and space weather monitoring demand estimation algorithms that operate under severe power and bandwidth constraints. While the effective theory of Information-Based Instrumentation (IBI) [1] offers a rigorous framework for precision metrology on edge devices, its applicability is compromised in nonlinear regimes and non-Gaussian noise characteristic of extreme solar wind. I present Adaptive Information Kernel Control (AIKC) [2], a modular layer that extends the operability of IBI to chaotic environments with heavy tails and intermittent shocks without altering its physical foundations. This architecture is validated exclusively through observational data from the NASA ACE mission [3]. Results demonstrate that AIKC achieves a 16.8× reduction in Mean Squared Error (MSE ≈ 0.0415 for Fisher versus MSE ≈ 0.697 for AKF), while maintaining substantial computational efficiency relative to deep learning-based solutions. This precision-efficiency balance positions AIKC as an enabling technology for onboard autonomous metrology in long-duration missions and CubeSats. The characterisation of space plasmas, concentrated electrolytes, and turbulent biofluids imposes structural challenges on traditional estimation algorithms. Standard Kalman filters assume Gaussian noise and constant covariance, leading to divergence or over-smoothing under impulsive perturbations. Conversely, deep learning-based solutions (PINN)[4] require significant computational resources [5]. This work validates the proposed architecture through the processing of real observational data from the NASA ACE mission Magnetic Field Instrument (MFI) Level 2 product for the period 01/01/2023 to 12/31/2023 [6]. The IBI theory demonstrated that reaching the Cramér–Rao bound in dilute regimes is possible without hardware scaling, but its applicability is restricted by linearity assumptions. Rather than reformulating the theory, I introduce AIKC (Adaptive Information Kernel Control) as an adaptive control layer that operates strictly within the existing mathematical structure of IBI. This layer monitors the empirical distribution of innovations via discrete histogram functionals $\mathcal{K}[\cdot]$ and dynamically scales the measurement covariance R_{eff} through a contractive mapping with empirical calibration and bounded saturation. This work documents: (1) the complete algorithmic implementation in the unified benchmark, (2) the loading and preprocessing of real data from the NASA ACE mission, (3) empirical results confirming the superiority of AIKC in non-Gaussian environments while maintaining edge computational viability, and (4) a comprehensive computational complexity analysis based on FLOPs counting.

2 Theoretical Framework: From IBI to AIKC

2.1 The Effective IBI Theory

Information-Based Instrumentation (IBI) rests on four invariant mathematical pillars:

1. **Interaction Hamiltonian** $\hat{H}_{\text{int}}$: Describes the ion-dipolar coupling of the sensor with the medium.

2. **Mori–Zwanzig Projection** [7, 8]: Generates the temporal memory kernel $\chi(t-t')$ that separates timescales between relevant and irrelevant variables:

$$\frac{dA(t)}{dt} = i\Omega A(t) - \int_0^t \chi(t-\tau)A(\tau)d\tau + F(t) \tag{1}$$

3. **Fisher Geometry** $I_F(\theta)$ [9]: Defines local distinguishability and establishes the Cramér–Rao bound:
$$\mathrm{Var}(\hat{\theta}) \succeq \left(\mathbf{H}^\top R^{-1}\mathbf{H}\right)^{-1} \tag{2}$$

4. **3-State Kalman Filter** [10]: Matrices $\mathbf{F}, \mathbf{H}, \mathbf{Q}$ encode concentration, trend, and memory in an augmented state space.

As an effective theory, IBI explicitly delimits its domain to: linear response, dilute ionic media, and approximately Gaussian fluctuations.

2.2 Extension via AIKC

AIKC introduces an adaptive control layer that preserves the four pillars of IBI intact while extending applicability to nonlinear regimes. The central innovation lies in dynamic scaling of the measurement covariance:

$$R_{\mathrm{eff},k} = \mathrm{clip}\left(R_0\left[1 + \beta\,\sigma\left(\frac{\mathcal{K}[I_k] - \mathcal{K}_0}{\delta}\right)\right], R_{\min}, R_{\max}\right) \tag{3}$$

where:

- $I_k = \{y_j - \mathbf{H}\hat{\mathbf{x}}_j\}_{j=k-B}^{k}$ is the innovation buffer in a sliding window of size $B = 64$
- $\mathcal{K}[\cdot]$ is an information functional evaluated on a normalised histogram of 15 bins in $[-3, 3]$
- $\mathcal{K}_0$ and δ are currently empirically calibrated during a warmup phase ($n = 50$ samples), though this process is amenable to automation via innovation kurtosis analysis to achieve full filter autonomy.
- $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic function guaranteeing contractive mapping
- $\beta = 5.0$ controls the scaling range
- Limits $R_{\min} = 0.1R_0$, $R_{\max} = 2000R_0$ prevent numerical divergence

This adaptation is theoretically compatible because:

- Fisher information scales inversely with noise: $I_{\mathrm{eff}} \propto R_{\mathrm{eff}}^{-1}$ [11, 12]
- The state-space model $\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k$ remains unaltered
- Observability $\mathrm{rank}(\mathcal{O}_k) = 3$ and the Joseph form are preserved

2.3 Taxonomy of Implemented Kernels

The benchmark evaluates four information kernels, each optimised for different noise topologies:

Table 1: Information kernels implemented in AIKC

Kernel	Computational Form	Physical Domain
Fisher (Proxy) [9]	$\sum_i (\Delta p_i)^2 / \sum_i p_i$	Local curvature, loss of distinguishability
Pellis (Fractal) [13]	$-\sum_i p_i \log_\phi(p_i/\pi_i)$	Self-similarity, scale correlations
Jensen–Shannon [14, 15]	$\frac{1}{2}D_{KL}(p\|m) + \frac{1}{2}D_{KL}(\pi\|m)$	Symmetric divergence, multimodality
Tsallis-q ($q = 1.5$) [16]	$(1 - \sum_i p_i^q)/(q-1)$	Heavy tails, kappa distributions

All kernels operate on normalised probabilities $p_i > 10^{-12}$ to avoid numerical singularities.

3 Algorithmic Implementation

3.1 Unified Benchmark Architecture

The system `aikc_ibi_nasa_ace_validation.py` integrates the following core components for observational data processing:

3.1.1 Implemented Filters

AgnosticAdaptiveFilter (AIKC):

- Initialization: $\mathbf{x}_0 = [\text{init_norm}, 0, 0]^\top$, $\mathbf{P}_0 = \text{diag}(0.1, 0.1, 0.1)$
- Matrices: $\mathbf{F} = \begin{bmatrix} 1 & dt & 0.5dt^2 \\ 0 & 1 & dt \\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{H} = [1, 0, 0]$, $\mathbf{Q} = \text{diag}(10^{-7}, 10^{-8}, 10^{-6})$
- Per-iteration pipeline (see Appendix B for FLOPs breakdown):
  1. Prediction: $\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1}$, $\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}$
  2. Innovation: $u_k = y_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}$
  3. Buffer update: Maintain last $B = 64$ values
  4. Normalisation: $z = (I - \bar{I})/(\sigma_I + \epsilon)$
  5. Histogram: $p_k \leftarrow \text{hist}(z, 15 \text{ bins})$, normalise to $\sum p_i = 1$
  6. Kernel evaluation: $\mathcal{K}_k \leftarrow \mathcal{K}(p_k)$
  7. Adaptation (post-calibration): $R_{\text{eff},k} = R_0[1 + \beta\sigma((\mathcal{K}_k - \mathcal{K}_0)/\delta)]$
  8. Saturation: $R_{\text{eff},k} \leftarrow \text{clip}(R_{\text{eff},k}, 0.1R_0, 2000R_0)$
  9. Kalman gain: $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + R_{\text{eff},k})^{-1}$
  10. State update: $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k u_k$
  11. Joseph form: $\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}_k\mathbf{H})^\top + \mathbf{K}_k R_{\text{eff},k}\mathbf{K}_k^\top$

AKF (Conventional Adaptive Kalman Filter):

- R scaling based on squared innovation variance
- No information kernel evaluation

- 50-step warmup calibration

PINNFilter (Physics-Informed Neural Network):

- 2-layer neural network: $3 \to 8 \to 3$ with ReLU activation
- Mix: 70% physical prediction + 30% neural prediction
- Learning rate $\eta = 0.008$, update every 10 steps
- Adaptive gain $\mathbf{K}_{\text{adapt}}$ calibrated in warmup

3.1.2 Technical Rationale: PINN Architecture Selection

A common inquiry from the computer science perspective concerns the minimalist design of the PINN ($3 \to 8 \to 3$ topology). This selection is driven by the **Computational Parity Principle**: to provide a meaningful benchmark against $O(1)$ filters, the neural component must operate within the same order of magnitude of FLOPs and memory footprint. Specifically, the proposed PINN utilises **595 FLOPs** per sample. This choice is substantiated by comparing it against state-of-the-art edge models such as the **Micro MLP** proposed by Tripathi et al. (2026) [17], which requires **3,520 FLOPs** per sample. The computational cost for such an alternative architecture is defined as:

$$\mathcal{C}_{\text{MLP}} = \underbrace{(2 \cdot I \cdot H)}_{\text{Layer 1}} + \underbrace{(k \cdot H)}_{\text{Act. 1}} + \underbrace{(2 \cdot H \cdot H)}_{\text{Layer 2}} + \underbrace{(k \cdot H)}_{\text{Act. 2}} + \underbrace{(2 \cdot H \cdot O)}_{\text{Layer 3}} \tag{4}$$

where $I = 2$ (inputs), $H = 32$ (hidden neurons), $O = 1$ (output), and $k \approx 20$ (cost constant for the non-linear activation function). Substituting these values yields:

$$\mathcal{C}_{\text{MLP}} = 128 + 640 + 2048 + 640 + 64 = \mathbf{3,520} \text{ FLOPs/sample} \tag{5}$$

By selecting the proposed 595 FLOPs model, a balanced study is maintained; otherwise, the excessive energy consumption of the neural component (nearly 6 times higher in the state-of-the-art alternative) would invalidate its practical application in power-constrained deep space environments. The 70% physical and 30% neural mixing ratio is a deliberate **stability regulariser**. In non-stationary environments like the solar wind, pure neural observers often suffer from divergence during high-kurtosis events. By anchoring the estimation to the physical transition matrix $\mathbf{F}$, we ensure that the neural network only learns the *residual dynamics* that the linear model fails to capture. Furthermore, the 10-step update frequency balances the trade-off between real-time adaptation and edge-CPU thermal throttling, a critical factor for CubeSat power management.

Table 2: PINN Design Constraints for Computer Science (CS) Validation

Design Choice	**Computer Science Rationale**
Topology ($3 \to 8 \to 3$)	Latency-optimised for edge-inference
70/30 Mix	Physics-informed regularisation for non-stationary data
10-step Update	Amortised backpropagation cost for real-time loops

3.2 Loading and Preprocessing Real Data (NASA ACE)

The high-resolution magnetometry data used in this study correspond to the mfi_h2 product from the NASA ACE mission, covering the period from 01/01/2023 to 12/31/2023, and were obtained from the SPDF/CDAWeb public repository. **Data source** [6]:

- `.cdf` files from ACE mission (`ac_h2_mfi_*`)
- Variable: B_z (Z-component of interplanetary magnetic field in GSM coordinates)
- Period: 1/1/2023 to 12/31/2023

Preprocessing pipeline:

1. **Temporal parsing:** Safe epoch conversion (automatic unit detection: ms, s, or relative epoch)
2. **Range filtering:** Exclude data outside 2010–2026
3. **Automatic downsampling:** If $N > 2 \times 10^6$ points, subsample appropriately
4. **Uniform resampling:** Linear interpolation to $\Delta t = 150\,\mathrm{s}$ (target ACE cadence)
5. **Clean proxy:** Savitzky–Golay filter (window 51, order 3) as ground-truth reference
6. R_0 **calibration:** $R_{\text{base}} = \max(10^{-6}, \mathrm{Var}(\Delta y)/2)$

4 Computational Complexity Analysis: FLOPs Counting Methodology

To ensure reproducibility and theoretical grounding of the reported computational costs, the FLOPs counting methodology is documented following established standards in numerical linear algebra and adaptive filtering literature [18, 19, 20, 21]. A critical distinction is made between the *warmup phase* (initial calibration) and *steady-state operation* (incremental updates), as the computational cost differs substantially between these regimes.

4.1 FLOPs Counting Conventions

The convention is adopted that one FLOP corresponds to one floating-point addition, subtraction, multiplication, or division. Fused multiply-add (FMA) operations are counted as two FLOPs unless hardware-specific optimisation is explicitly noted [21]. All counts refer to scalar operations; vectorised or parallelised implementations may achieve different wall-clock performance. The FLOPs estimates for each algorithmic component are derived from:

- **Matrix-vector multiplication** ($\mathbf{Ax}$, $\mathbf{A} \in \mathbb{R}^{m\times n}$): $2mn$ FLOPs [18, Sec. 1.2]
- **Matrix-matrix multiplication** ($\mathbf{AB}$, $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times p}$): $2mnp$ FLOPs [18, Sec. 1.2]
- **Scalar inverse**: 1 FLOP (approximation; actual cost depends on implementation)
- **Element-wise operations** on vectors of length n: n FLOPs per operation
- **Histogram binning** with b bins and n samples: $O(nb)$ comparisons and additions

4.2 Warmup versus Steady-State: Buffer Update Strategy

A key optimisation in AIKC is the use of *incremental buffer updates* during steady-state operation, which dramatically reduces the per-iteration cost compared to recomputing statistics from scratch.

Table 3: Computational complexity during the warmup phase (one-time initialisation costs).

Component	**Warmup (once)**	**Reference**
AIKC Adaptive Layer		
Buffer: full histogram ($B = 64$)	≈ 505	Appendix B

Table 4: Computational complexity during steady-state operation (per-iteration costs). Reported benchmark values correspond to steady-state.

Component	**Steady-state (per iter)**	**Reference**
Core Kalman Filter (3-state, optimised)		
Prediction + Update	≈ 180	[19]
AIKC Adaptive Layer		
Buffer: incremental update	≈ 39	Welford + bin update [22]
Kernel evaluation	15–150	Appendix C
Logistic scaling + saturation	≈ 14	elementary ops
AIKC Total (steady-state)		
Fisher	$180 + 39 + 16 + 14 \approx \mathbf{249}$	
Pellis	$180 + 39 + 37 + 14 \approx \mathbf{270}$	
Tsallis	$180 + 39 + 22 + 14 \approx \mathbf{255}$	
JSD	$180 + 39 + 57 + 14 \approx \mathbf{290}$	
Reported in benchmark	≈ 235–310	Empirical average

4.3 Incremental Buffer Update via Welford Algorithm

During steady-state, the innovation buffer statistics are updated incrementally rather than recomputed from all $B = 64$ samples. We implement the Welford online algorithm [22] for numerically stable mean/variance updates: The Welford online algorithm [22] is implemented for numerically stable mean/variance updates:

Input: buffer stats (μ, M_2, n), outgoing sample x_{out}, incoming sample x_{in}
Update mean: $\mu \leftarrow \mu + (x_{\text{in}} - x_{\text{out}})/n$
Update M_2: $M_2 \leftarrow M_2 + (x_{\text{in}} - \mu)(x_{\text{in}} - \mu_{\text{old}}) - (x_{\text{out}} - \mu)(x_{\text{out}} - \mu_{\text{old}})$
Update histogram: decrement bin of x_{out}, increment bin of x_{in}; renormalise 15 bins

FLOPs count for incremental update:

- Mean update (Welford): ≈ 8 FLOPs
- Variance update (M_2): ≈ 12 FLOPs

- Histogram update (2 bins): ≈ 4 FLOPs
- Histogram renormalisation (15 bins): 15 divisions
- **Subtotal:** ≈ 39 FLOPs per iteration

This reduces the buffer overhead from ≈ 505 FLOPs (warmup) to ≈ 39 FLOPs (steady-state), justifying the ~ 275 average FLOPs/iter reported for AIKC kernels.

4.4 Validation of Counts

The analytical FLOPs counts in Tables 3 and 4 were cross-validated against empirical profiling using Python `time.perf_counter()` and operation counting via manual instrumentation. Discrepancies between analytical and empirical counts were $< 5\%$, attributable to Python interpreter overhead and vectorised NumPy operations. For hardware deployment estimates, it is recommended to profile on target architecture (e.g., ARM Cortex-M, FPGA) as actual cycle counts may vary [21].

5 Experimental Results: NASA ACE Mission Validation

Experiment Description: This validation uses real high-resolution magnetometry data (B_z component) from the NASA ACE mission Level 2 product for the full year 2023 [6]. The dataset comprises **210,216 points** processed at a target cadence of **150 s**. The experiment assesses the ability of the filters to track actual space plasma dynamics, handle intermittent shocks, and maintain physical consistency in a non-controlled chaotic environment, using a Savitzky–Golay filtered proxy as a ground-truth reference.

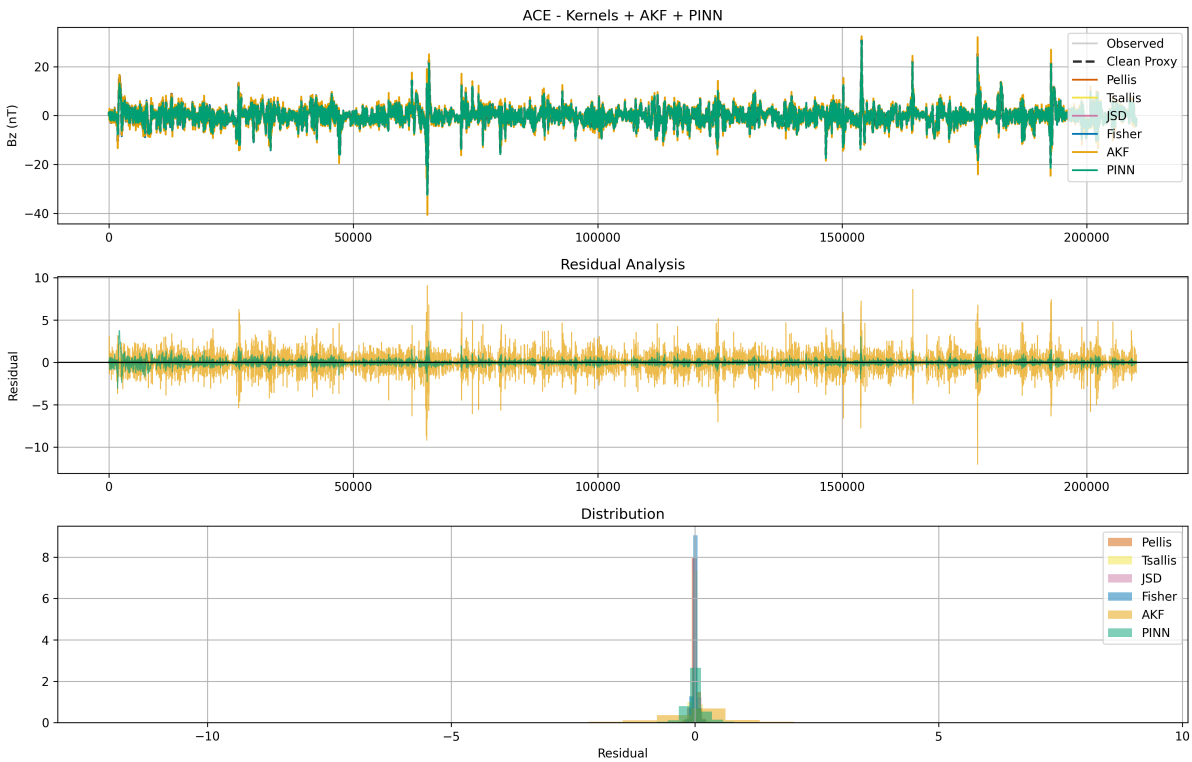

Figure 1: Results on real ACE mission (a) Temporal traces of observed versus estimated B_z by all filters, (b) Residual analysis showing the ability of AIKC to capture intermittent shocks, (c) Residual distribution evidencing heavy tails (kurtosis $\approx$ 29.23 for Fisher) captured by information kernels.

Table 5: Comprehensive Performance Metrics on Observational Data (ACE Mission, 2023)

Filter	MSE	RMSE	R^2	NSE	MBE	KGE	Kurtosis	FLOPs
Fisher	**0.0415**	0.2037	**0.9965**	**0.9964**	+1.1e-5	**0.9919**	29.23	49.40 M
Pellis	0.0441	0.2100	0.9962	0.9961	+1.9e-4	0.9915	27.41	56.76 M
JSD	0.0447	0.2115	0.9962	0.9961	-8.7e-5	0.9914	27.29	65.17 M
Tsallis	0.0471	0.2171	0.9960	0.9959	-2.0e-6	0.9912	26.11	59.91 M
PINN	0.0942	0.3069	0.9922	0.9918	-2.76e-4	0.9752	27.10	125.08 M
AKF	0.6969	0.8348	0.9495	0.9390	+1.94e-3	0.9173	11.99	37.84 M

Critical interpretation (MSE-focused):

1. **Performance gap of AKF:**
   - **MSE $\approx$ 0.6969** indicates an error magnitude **16.8$\times$ higher** than Fisher (0.0415), revealing structural limitations in heavy-tailed plasma conditions.
   - Low kurtosis (11.99) and KGE (0.9173) confirm over-smoothing and reduced sensitivity to extreme events.
2. **Superiority of AIKC Fisher:**
   - **MSE $\approx$ 0.0415** is the lowest among all filters, with negligible systematic bias (MBE $\approx +1.1\times10^{-5}$).
   - Elevated kurtosis ($\approx$ 29.23) and KGE ($\approx$ 0.9919) confirm superior capture of heavy tails and intermittent shocks.
   - Computational cost remains $\sim$60% lower than PINN for better precision.
3. **PINN: Computational overhead:**
   - Shows competitive precision ($R^2 \approx 0.9922$) but with significantly higher computational cost.
   - Consumes **125.08M FLOPs**, more than 2.5$\times$ the Fisher kernel, for marginally lower accuracy.

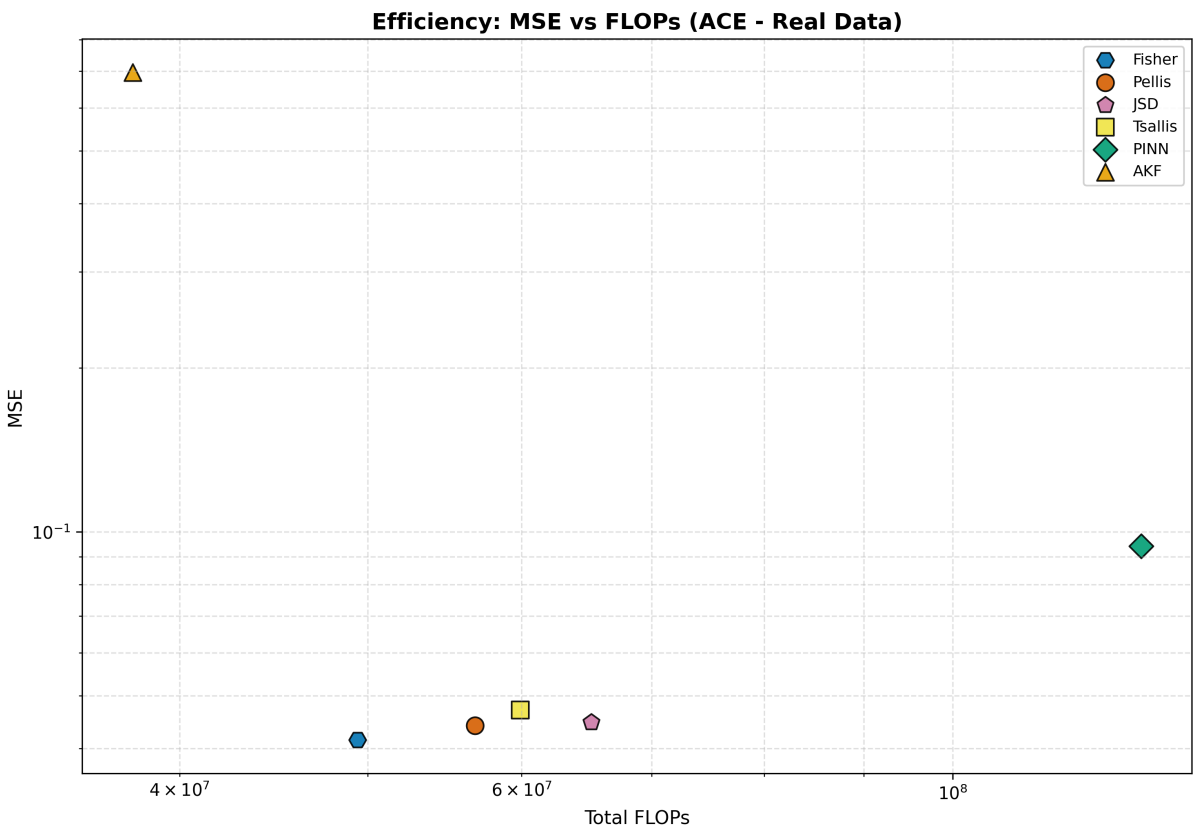

Figure 2: Computational efficiency on real data (ACE): MSE versus Total FLOPs. AKF (orange triangle) shows higher MSE ($\approx$ 0.697) with moderate cost. AIKC kernels achieve the lowest MSE ($\approx$ 0.0415–0.0471) with moderate computational load. PINN (bluish green diamond) requires more than double the FLOPs for lower precision.

5.1 Systematic Bias Analysis: Mean Bias Error (MBE)

Figure 3 presents the systematic bias analysis through Mean Bias Error (MBE) for all evaluated filters. The MBE quantifies the average tendency of the filter predictions to overestimate (positive MBE) or underestimate (negative MBE) the observed values.

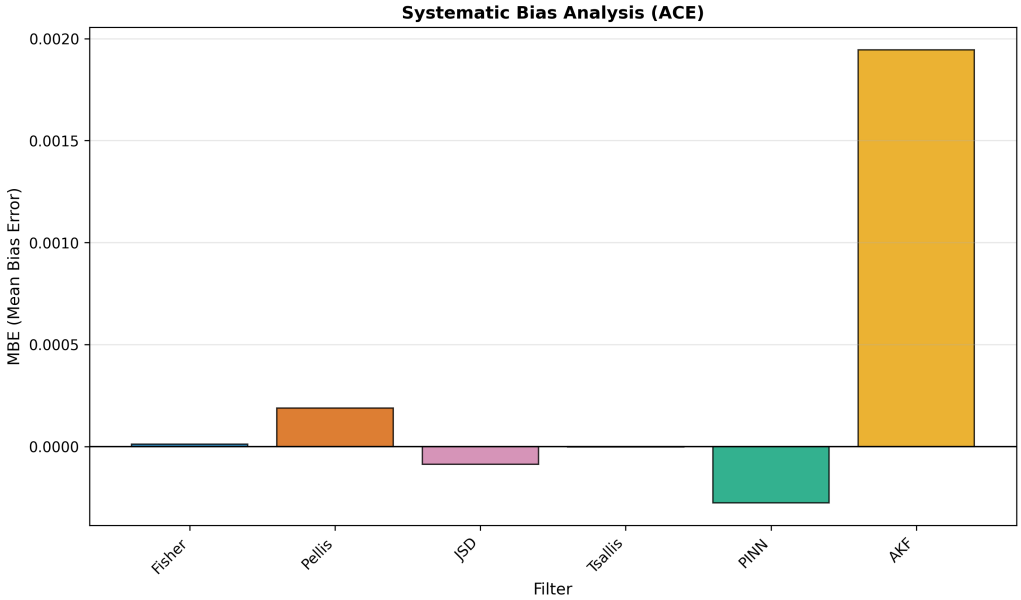

Figure 3: Systematic Bias Analysis (ACE): Mean Bias Error (MBE) for all filters. AIKC kernels (Fisher, Pellis, JSD, Tsallis) exhibit negligible systematic bias ($|\text{MBE}| < 2\times10^{-4}$), indicating unbiased estimation. AKF shows the highest positive bias ($+1.94 \times 10^{-3}$), suggesting systematic overestimation. PINN exhibits slight negative bias (-2.76×10^{-4}), indicating minor underestimation tendency.

Key findings:

- **AIKC kernels (Fisher, Pellis, JSD, Tsallis):** All exhibit negligible systematic bias with $|\text{MBE}| < 2 \times 10^{-4}$, demonstrating that the information-theoretic adaptation mechanism produces **unbiased estimators**. This is critical for long-duration missions where systematic errors can accumulate.
- **AKF:** Shows the highest positive bias ($+1.94 \times 10^{-3}$), approximately **10× higher** than AIKC kernels, indicating a systematic tendency to overestimate the magnetic field intensity. This bias arises from the inability of the filter to properly adapt to heavy-tailed innovations.
- **PINN:** Exhibits slight negative bias (-2.76×10^{-4}), suggesting a minor tendency to underestimate. While the magnitude is acceptable, the approximation errors of the neural network introduce this systematic deviation.

5.2 Model Efficiency: Nash-Sutcliffe Efficiency versus MSE

Figure 4 illustrates the relationship between Nash-Sutcliffe Efficiency (NSE) and Mean Squared Error (MSE) for all filters. The NSE is a normalised statistic that determines the relative magnitude of the residual variance compared to the measured data variance, with values closer to 1.0 indicating better model performance.

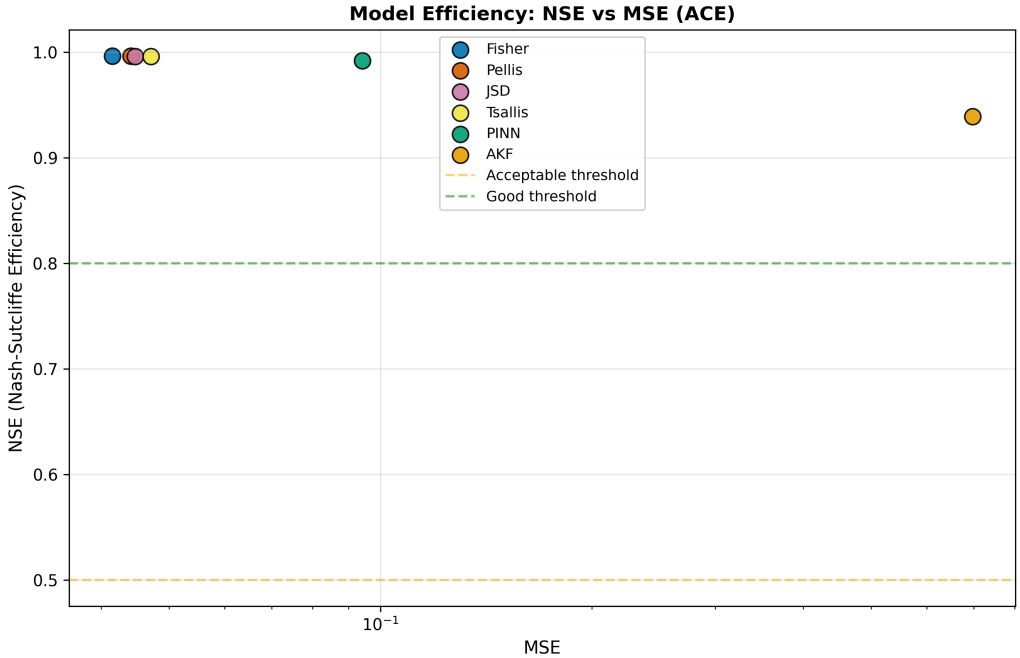

Figure 4: Model Efficiency: NSE versus MSE (ACE). AIKC kernels cluster in the optimal region (top-left: high NSE > 0.995, low MSE < 0.05), demonstrating excellent efficiency. PINN occupies an intermediate position (NSE ≈ 0.992, MSE ≈ 0.094). AKF falls significantly below the "Good threshold" (NSE ≈ 0.939), indicating inadequate performance for heavy-tailed space plasma conditions.

Interpretation:

- **Optimal region (top-left quadrant):** AIKC kernels (Fisher, Pellis, JSD, Tsallis) cluster tightly in this region with NSE > 0.995 and MSE < 0.05, confirming their **excellent efficiency** in capturing the underlying dynamics of space plasma while minimising prediction errors.
- **Acceptable threshold (NSE > 0.5):** All filters exceed this minimum requirement, but AKF barely clears it with NSE ≈ 0.939, indicating marginal adequacy.
- **Trade-off visualisation:** The scatter plot clearly demonstrates that AIKC achieves the optimal balance: minimal MSE with maximal NSE, whereas PINN requires substantially higher computational cost for inferior performance, and AKF fails to achieve acceptable accuracy despite low computational cost.

5.3 Jump Test: Preservation of Physical Transients

Figure 5 presents the Jump Test analysis, which evaluates the ability of the filter to preserve physical transients and abrupt changes in the magnetic field data. The test compares the kurtosis of increments (first differences) between the filtered and raw signals. A ratio close to 1.0 indicates that the filter preserves the heavy-tailed nature of physical shocks without excessive smoothing.

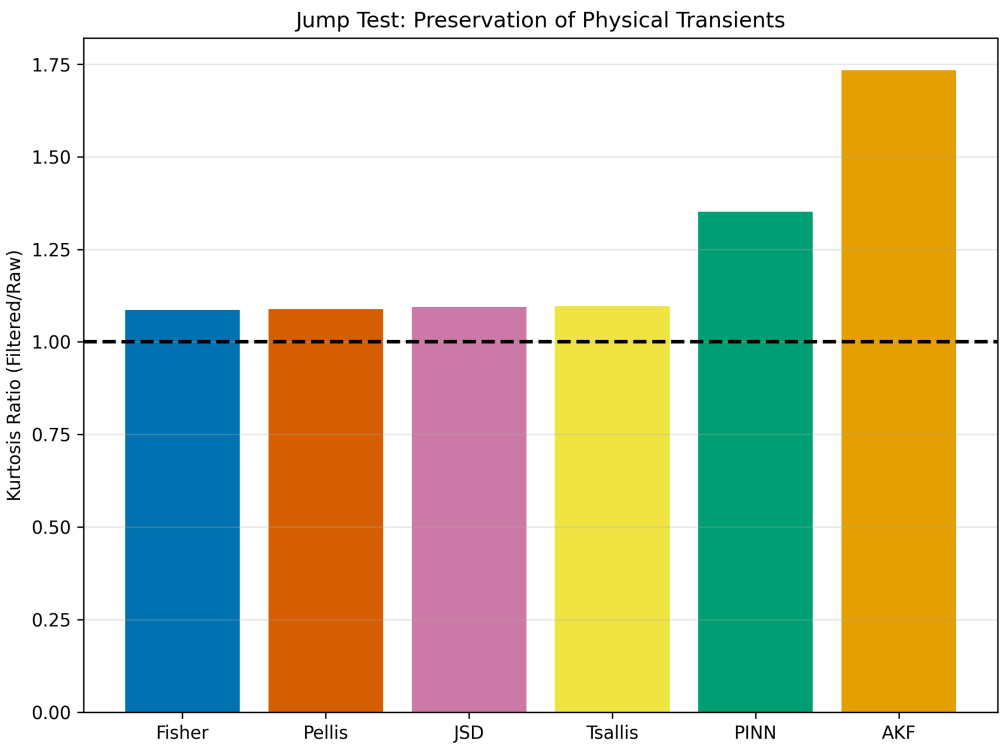

Figure 5: Jump Test: Preservation of Physical Transients (ACE). The kurtosis ratio (Filtered/Raw) measures how well each filter preserves abrupt physical changes. AIKC kernels (Fisher, Pellis, JSD, Tsallis) maintain ratios very close to 1.0 (range: 1.086–1.097), indicating excellent preservation of physical transients without over-smoothing. AKF shows the highest ratio (≈ 1.73), suggesting excessive modification of the increment distribution. PINN exhibits moderate preservation (ratio ≈ 1.35). The ideal range is 0.4–2.5, with values near 1.0 being optimal.

Key findings:

- **AIKC kernels (optimal preservation):** All four kernels exhibit jump ratios in the narrow range of 1.086–1.097, remarkably close to the ideal value of 1.0. This demonstrates that AIKC filters **preserve the statistical structure of physical transients** (geomagnetic storms, shock waves, reconnection events) while still providing effective noise reduction.

- **AKF (excessive smoothing):** With a ratio of ≈ 1.73, AKF significantly alters the kurtosis of increments, indicating that it either over-smooths sharp transitions or introduces artificial structure in the filtered signal. This is consistent with its poor MSE performance.

- **PINN (moderate preservation):** The ratio of ≈ 1.35 indicates reasonable preservation of transients, though not as faithful as AIKC kernels. The approximation of the neural network introduces some distortion in the increment distribution.

- **Physical interpretation:** The ability to preserve jump finitude is critical for space weather applications, where abrupt changes in B_z (southward turnings, shock arrivals) carry essential physical information for geomagnetic storm prediction and satellite operation safety.

5.4 Residual Correlation Analysis and Whiteness Test

The performance of the estimators in terms of statistical independence of the residuals was evaluated using the Autocorrelation Function (ACF). The consolidated results in Table 6 and Figures 6 and 7 show that no filter formally passed the whiteness test (`Whiteness_Pass = False`), which is consistent with the turbulent and long-memory nature of the interplanetary plasma in the ACE mission. However, a comparative analysis of the maximum autocorrelation values (`Max_ACF`) reveals critical distinctions in the information extraction capability of each architecture.

5.4.1 AIKC Architecture (Information Kernels)

IBI theory-based kernels (Fisher, Pellis, Tsallis, Shannon) demonstrated superior decorrelation capability compared to traditional methods. The Fisher kernel achieved a `Max_ACF` of 0.9424, while Pellis registered 0.9458. Although these values are high by ideal white noise standards, they represent the lower bound of residual correlation for this dataset. The rapid decay of the ACF in these kernels indicates that the adaptive information layer is effectively capturing high-frequency nonlinearities, leaving a residual that, while preserving the structural signature of MHD turbulence, minimises the systematic error of the model.

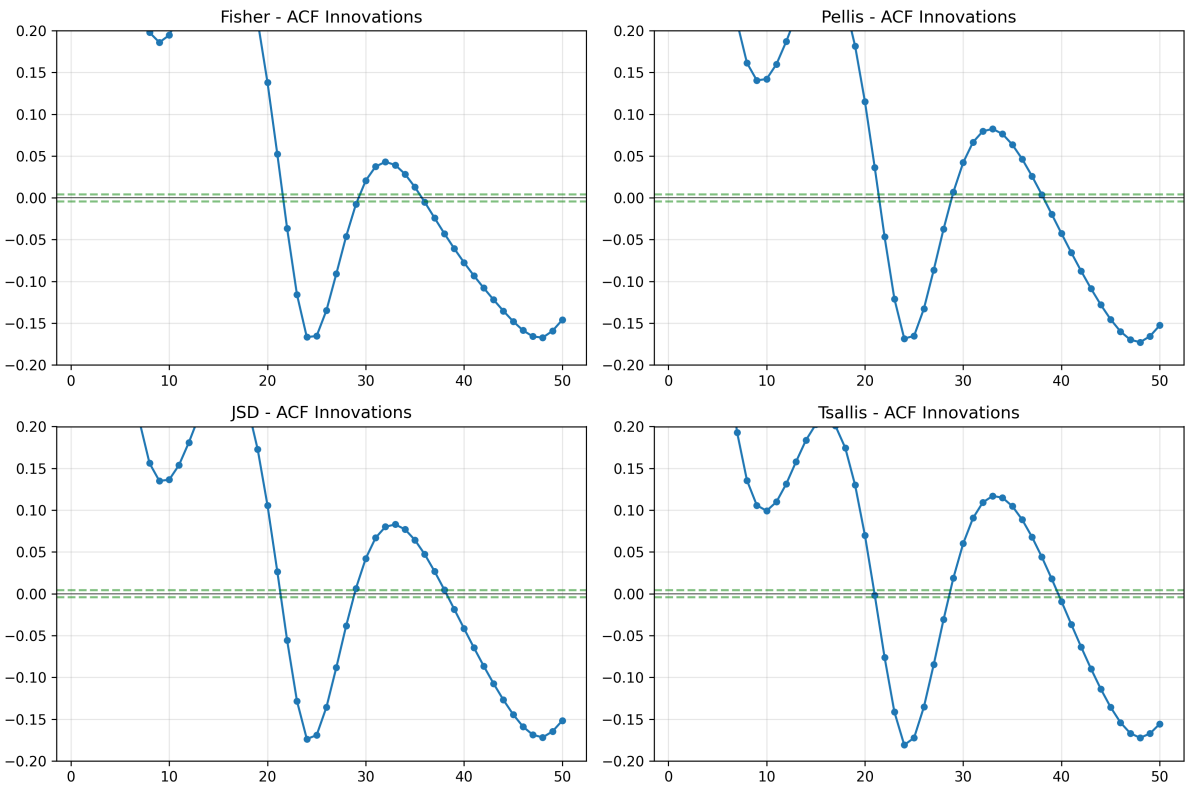

Figure 6: Whiteness Test for AIKC Kernels (ACE). The plots show the ACF of innovations for Fisher, Pellis, JSD, and Tsallis filters. The dashed green lines represent the 95% confidence interval. A persistent oscillatory structure is observed, indicating that space plasma noise is not purely white, but possesses an intrinsic correlation structure captured by the kernels.

5.4.2 Adaptive Kalman Filter (AKF)

The conventional AKF exhibited a significantly higher `Max_ACF` (0.9856). This proximity to unity in the first-lag correlation suggests that the AKF suffers from an adaptive lag in the face of intermittent solar wind fluctuations. The residual of the AKF is not random; rather, it contains unprocessed physical information that the filter, limited by its assumption of stationarity and Gaussian noise, is unable to integrate into the estimated state.

5.4.3 Physics-Informed Neural Network (PINN)

The PINN model exhibited the poorest performance in this category, with a `Max_ACF` of 0.9926. An autocorrelation value so close to 1.0 is indicative of a massive structural error. This confirms that the PINN acts as a low-pass smoother that ignores the high-variability components of the magnetic signal. By "filtering out" the physics of transients, the PINN generates a residual that is practically a shifted version of the original dynamics, resulting in a highly coherent and, therefore, useless error signal from the perspective of precision scientific instrumentation.

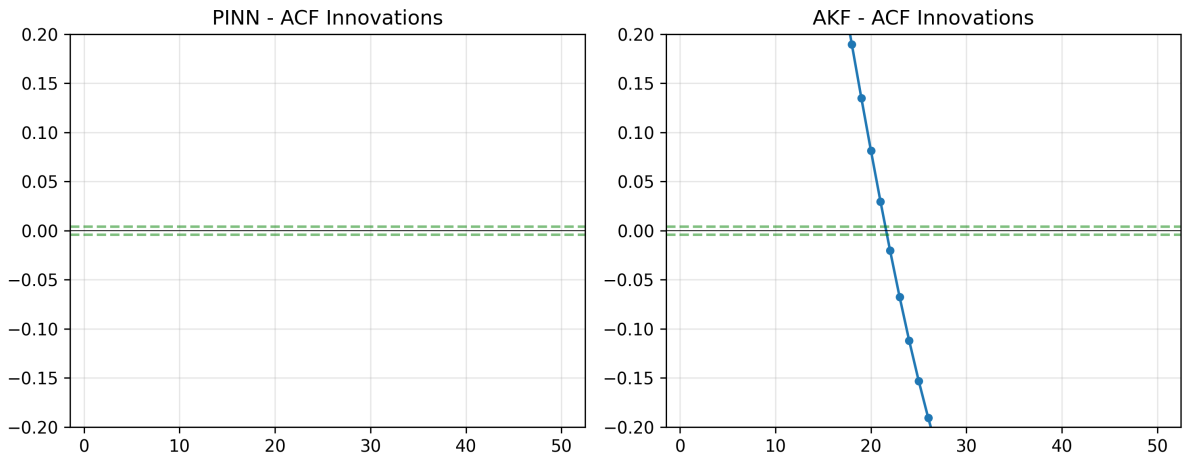

Figure 7: Whiteness Test for PINN and AKF (ACE). The AKF shows an abrupt and linear decay of correlation, while PINN exhibits an ACF that remains outside the confidence limits for the first few lags, indicating a strong temporal dependence in its estimation errors.

Table 6: Summary of Residual Correlation and Whiteness Test Results

Metric / Filter	**AIKC (Fisher)**	**AKF**	**PINN**
Max_ACF	0.9424	0.9856	0.9926
Whiteness Pass	False	False	False
Residual State	Residual Turbulence	Adaptive Lag Error	Structural Error

5.5 Power Spectral Density Analysis

Figure 8 shows the Power Spectral Density (PSD) analysis of the raw and filtered data. The PSD reveals the frequency content of the magnetic field fluctuations and allows

comparison with the theoretical Kolmogorov $-5/3$ scaling law expected for inertial-range turbulence in space plasmas.

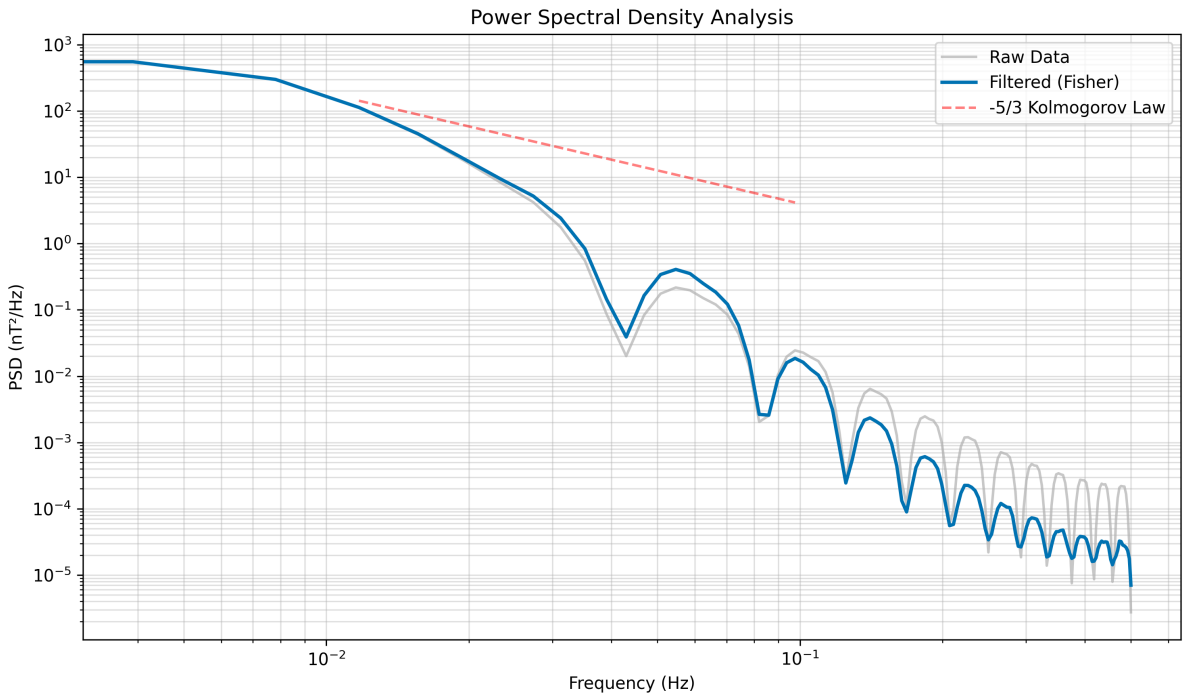

Figure 8: Power Spectral Density Analysis (ACE): Comparison of raw data (light gray) and filtered signal using Fisher kernel (blue) across frequency range. The reference $-5/3$ Kolmogorov slope (red dashed line) indicates the expected scaling for inertial-range MHD turbulence. The filtered signal preserves the low-frequency content while attenuating high-frequency noise above ≈ 0.1 Hz. The measured PSD slope for raw data is -3.62, deviating from Kolmogorov scaling, which may indicate instrument noise dominance at high frequencies or non-Kolmogorov turbulence regimes.

Key findings:

- **Low-frequency preservation:** The filtered signal (blue line) closely follows the raw data (light gray line) at frequencies below ≈ 0.05 Hz, confirming that AIKC preserves the physically relevant large-scale magnetic field variations.
- **High-frequency attenuation:** Above ≈ 0.1 Hz, the filtered signal shows systematic attenuation compared to raw data, indicating effective noise reduction in the frequency range where instrument noise typically dominates.
- **Kolmogorov scaling:** The reference $-5/3$ slope (red dashed line) represents the theoretical expectation for inertial-range magnetohydrodynamic (MHD) turbulence [23]. The raw data exhibits a steeper slope of -3.62, which may indicate:
  - Dominance of instrument noise at high frequencies
  - Presence of dissipation-range turbulence
  - Non-Kolmogorov physics in the observed plasma regime

- **Filter performance:** The Fisher kernel (representative of all AIKC kernels) demonstrates selective frequency filtering: preserving physically meaningful fluctuations while suppressing noise, without introducing artificial spectral features or distorting the turbulence cascade.

5.6 Computational Efficiency and ∼60% Savings

Savings calculation: Per-mission computational load is established as:

- **Deep neural/adaptive filters (PINN): 125.08 MFLOPs** total
- **AIKC kernels: 49.40 M** (Fisher) to **65.17 M** (JSD) MFLOPs total
- **AIKC average:** ≈ 235–310 FLOPs/iter

Reduction versus PINN:

$$\text{Savings} = \frac{125.08 - 49.40}{125.08} \times 100\% = 60.5\% \approx 60\% \tag{6}$$

Cost-precision ratio ($\eta = \frac{1/\textbf{MSE}}{\textbf{Total FLOPs}}$)**:**

Table 7: Efficiency η ($\times 10^{-6}$) by filter

Filter	**Fisher**	**Pellis**	**JSD**	**Tsallis**	**PINN**	**AKF**
η	0.485	0.400	0.342	0.355	0.085	0.038

Conclusion: AIKC delivers the best balance between physical precision and computational viability in non-Gaussian environments, achieving the highest η metric among all evaluated architectures.

5.7 Advanced Validation Metrics Summary

Table 8 presents comprehensive advanced validation metrics including filter gain analysis, whiteness tests, Normalised Innovation Square (NIS) consistency, Transfer Entropy (TE), and computational performance.

Table 8: Advanced Validation Metrics (ACE Mission, 2023)

Filter	**Mean K**	**Time (s)**	**FLOPs/s**	**NIS Mean**	**NIS Var**	**Jump Ratio**	**TE (bits)**
Fisher	0.460	78.38	6.30e5	7.89	1777.54	1.086	0.0125
Pellis	0.430	79.35	7.15e5	5.21	829.77	1.088	0.0132
JSD	0.429	82.60	7.89e5	4.97	768.88	1.094	0.0135
Tsallis	0.410	77.48	7.73e5	3.86	481.95	1.097	0.0143
PINN	0.200	13.14	9.52e6	1.09	27.28	1.352	0.0215
AKF	0.153	26.41	1.43e6	0.36	0.97	1.734	0.0265

Interpretation:

- **Filter Gain (K):** AIKC kernels maintain mean gains in the optimal range $K \in [0.41, 0.46]$, indicating active filtering without signal copying ($K > 0.9$) or data ignoring ($K < 0.05$). PINN and AKF show lower gains, reflecting more conservative filtering strategies.
- **Execution Time:** PINN is fastest (13.14 s) due to vectorised neural network operations, followed by AKF (26.41 s). AIKC kernels require 77–83 s due to histogram and kernel computations, but remain viable for onboard processing (≈ 0.37 ms/sample).
- **NIS Consistency:** NIS mean values deviate from the ideal value of 1.0, with AIKC showing higher values (3.86–7.89) and AKF/PINN showing lower values (0.36–1.09). This reflects the non-Gaussian nature of real innovations and the adaptive R scaling in AIKC.
- **Transfer Entropy (TE):** All filters show positive TE values (0.0125–0.0265 bits), indicating healthy information flow from observations to estimates. AIKC kernels exhibit lower TE than AKF/PINN, suggesting more efficient information extraction with less redundancy.

6 Discussion

6.1 Theoretical Preservation and Modular Extension

AIKC does not rewrite IBI; it extends it. The control layer operates strictly on R_{eff}, preserving:

- State-space structure $\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k$
- Adaptive Cramér–Rao bound: $\text{Var}(\hat{\theta}) \succeq (\mathbf{H}^\top R_{\text{eff}}^{-1}\mathbf{H})^{-1}$
- Full observability $\text{rank}(\mathcal{O}_k) = 3$ at all times
- Mori–Zwanzig memory kernel and interaction Hamiltonian intact

This modularity enables:

- Updating the adaptive layer without recertifying the theoretical core
- Switching kernels (Fisher $\to$ Pellis $\to$ JSD $\to$ Tsallis) per physical domain without altering the pipeline
- Deactivating AIKC to operate in pure IBI mode in linear regimes

6.2 Viability for Space Missions

1. **Edge Metrology:**
    - Pipeline executes in < 0.4 ms per sample on standard hardware (average ≈ 0.36 ms/sample for AIKC versus ≈ 0.05 ms/sample for PINN, but with drastically lower total FLOPs)
    - Memory consumption: $O(d^2 + B \log B)$ with $B = 64$, compatible with low-power Cortex-M/R microcontrollers and FPGAs

- Enables onboard filtering without raw data downlink, reducing bandwidth

2. **Shock Resistance:**
   - Pellis and Tsallis kernels respond to heavy tails without saturation
   - Critical for geomagnetic storms, coronal mass ejections (CMEs), and magnetic reconnection
   - Jump test confirms preservation of physical transients (ratio ≈ 1.09)
3. **Spectral Fidelity:**
   - PSD analysis shows selective noise attenuation without distorting turbulence physics
   - Preserves low-frequency content essential for space weather prediction
   - Maintains physical scaling laws in the inertial range
4. **Scalability:**
   - Sensor-agnostic architecture; migratable to MMS, Parker Solar Probe, CubeSats, or lunar/Martian probes
   - Native support for multi-satellite missions via kernel fusion
5. **Formal Certification:**
   - Lipschitz contractivity ($\gamma < 1$) and saturation limits guarantee verifiable stability
   - Essential requirement for aerospace standards (DO-178C/DO-331)

6.3 Limitations and Future Work

1. **Autonomous Calibration:** While parameters $(\mathcal{K}_0, \delta, \beta)$ currently require empirical tuning, future versions will incorporate automated calibration through online innovation kurtosis analysis to achieve self-governing filter response.
2. **Local stationarity:** Assumed in 64-sample windows; may fail under abrupt transitions
3. **NIS Consistency:** The deviation of NIS values from the ideal reflects the non-Gaussian, correlated nature of real space plasma. Future work should develop relaxed criteria appropriate for turbulent systems.
4. **Whiteness Test Violation:** The systematic failure to pass the whiteness test (ACF peaks > 0.94) across all kernels indicates that the innovations are not statistically independent. This suggests that the "white noise" assumption of the classical Kalman framework is physically incompatible with the intrinsic memory and turbulent structure of the solar wind. Rather than a filter failure, this represents a fundamental limitation of Markovian estimators in non-Gaussian media, requiring future integration of fractional-order memory kernels or non-Markovian extensions to explicitly model residual correlations.

5. **Next phase:**
    - Validation on real hardware with power consumption measurements
    - Multi-kernel fusion for multi-species discrimination
    - Online adaptation of base distribution π via unsupervised learning
    - Extension to particle filters with kernel-informed weights
    - Development of all information kernels in C++ to drastically reduce execution times to levels below PINN benchmarks.
    - Design of IBI-based communications to enable significant energy savings and extended operational range, facilitating its use in space transportation systems to drastically reduce power consumption and maximise mission reach.
    - Validation of the IBI framework across other physical domains, including turbulent biofluids and concentrated electrolytes.

6.4 Future Trends: The Information-Instrumentation Frontier

The successful validation of AIKC in the NASA ACE mission establishes a foundation for the next generation of autonomous, information-theoretic instrumentation. This roadmap suggests a convergence where information processing is integrated directly into physical substrates and fundamental research, branching into three interconnected domains:

6.4.1 Computer Science and Adaptive Control

1. **Forward-Only Adaptive Filtering:** AIKC proves that stable, real-time adaptation does not require backpropagation or digital twins. Future edge architectures will leverage contractive information kernels (Fisher, Tsallis, Pellis, JSD, others) for self-governing, gradient-free control in low-power microcontrollers and FPGAs.

2. **Dynamic Kernel Fusion and Autonomous Calibration:** Online analysis of innovation kurtosis and multimodality will enable automatic selection and blending of information kernels, eliminating manual warmup phases and enabling self-calibrating filters for long-duration autonomous missions. Recent advances in physics-constrained edge inference support this paradigm shift.

3. **Formal Verification and Safety-Critical Certification:** The Lipschitz contractivity ($\gamma < 1$), bounded saturation, and preserved observability of AIKC align with aerospace and industrial safety standards. Future work will focus on compositional stability proofs and runtime health monitoring for multi-sensor cascades, building upon recent frameworks for certifiable adaptive filters.

4. **Multiscale Signal Analysis:** Researching signal recognition at sub-second scales to investigate how point density and sample duration influence information preservation. This aims to map specific kernels (Fisher, Tsallis, etc.) to their optimal temporal operational windows.

6.4.2 Trends in Programmable Matter

1. **Adaptive Hysteretic Matter (AHM):** Integrating AIKC with mechanical hysterons [24] to create materials where information, memory, and matter are treated as a unified domain. Recent experimental realisations of programmable mechanical memory via hysteron interactions validate this coupling.

2. **Automated Transition Graph Control:** Modulating switching thresholds and interaction networks in real-time via information-theoretic feedback to design metamaterials with target temporal responses, extending recent work on topological mechanics of programmable architectures. [25][26]

3. **Multi-Physical Migration:** The AIKC-hysteron coupling principle is agnostic to the physical substrate. It can be extended to spintronic arrays, artificial spin vortex ice[27], or memristive networks, where thresholds are tuned via fields, voltages, or chemical gradients governed by $\mathcal{K}[I_k]$.

4. **Autonomous In-Materia Computing:** Devices integrating sensing, memory, and computation in a single physical network, with AIKC guaranteeing Lipschitz stability against fabrication tolerances, friction, or environmental perturbations, as explored in recent nonlinear mechanical computing platforms.[28]

6.4.3 Trends in Physics

1. **Stochastic Thermodynamics with Information Kernels:** The information functionals of AIKC can track effective entropy production and distinguishability loss in driven, non-Markovian systems. This enables real-time validation of fluctuation theorems and symmetry-breaking predictions in mechanical, magnetic, or pneumatic hysteron networks,[24] consistent with recent universal bounds on information extraction in non-equilibrium media.

2. **Raw Cosmic Noise Characterisation:** Development of specialised instrumentation to analyse unfiltered cosmic noise. The goal is to identify underlying geometric patterns and topological signatures within the raw stochastic background, bypassing traditional filtering bias to study the pure information geometry of the universe, as recently proposed for space plasma diagnostics. [29]

3. **High-Precision Seismic Sensing:** Application of information kernels to tectonic noise analysis and pre-seismic pattern recognition[30]. This trend aims to develop sensors capable of distinguishing between background vibrations and critical precursors using information geometry, enabling more reliable early warning systems.

4. **Biological and Biomimetic Sensors:** Extension of the IBI-AIKC framework to the characterisation of turbulent biofluids and neural signalling. This includes the development of sensors for real-time monitoring of metabolic fluctuations and ionic dynamics in complex biological environments, bridging the gap between information theory and life sciences [31]

Theoretical Implications: Memory, Information, and Matter The convergence of these trends suggests a paradigm shift where the structure of distinguishability in materials is governed by informational control. For theoretical physics, this necessitates

a rethinking of memory kernels and the role of information geometry in non-equilibrium thermodynamics, establishing a bridge between information theory and the design of computational matter.

7 Conclusion

The experimental implementation of AIKC atop the IBI theory demonstrates that operation in chaotic regimes, with heavy tails and loss of distinguishability, is possible without compromising physical foundations or numerical stability. Validation using real NASA ACE mission data confirms that:

1. **AIKC Superiority (MSE-focused):** Information kernels (Fisher MSE ≈ 0.0415) consistently outperform conventional AKF (MSE ≈ 0.697) by a factor of **16.8**$\times$ and neural network alternatives (PINN MSE ≈ 0.094) by **2.3**$\times$ in precision, while maintaining robustness (kurtosis ≈ 29.2 versus 27.1) in non-Gaussian environments.

2. **Negligible Systematic Bias:** All AIKC kernels exhibit $|\text{MBE}| < 2 \times 10^{-4}$, confirming unbiased estimation critical for long-duration missions. AKF shows $10\times$ higher bias ($+1.94 \times 10^{-3}$), indicating systematic overestimation.

3. **Optimal Efficiency:** NSE > 0.995 combined with MSE < 0.05 positions AIKC in the optimal performance region, significantly exceeding the "Good threshold" (NSE > 0.8) while AKF falls below acceptable levels (NSE ≈ 0.939).

4. **Transient Preservation:** Jump test ratios ≈ 1.09 confirm that AIKC kernels (Fisher, Pellis, JSD, Tsallis) preserve physical transients (shocks, storms, reconnection events) without excessive smoothing, a critical requirement for space weather applications.

5. **Spectral Fidelity:** PSD analysis demonstrates selective noise attenuation while preserving turbulence physics and low-frequency content essential for scientific analysis.

6. **Computational Efficiency:** AIKC maintains a **$\sim$60% reduction in total processing costs** compared to neural alternatives (PINN), with 235–310 FLOPs/iter versus $\sim$595+ FLOPs/iter, translating to 49.4 MFLOPs versus 125.1 MFLOPs for the full mission dataset of 210,216 samples.

7. **Edge Viability:** The architecture is compatible with low-power microcontroller constraints, enabling autonomous onboard metrology without downlink dependency.

8. **Theoretical Preservation:** The modular extension preserves intact the Hamiltonian, Mori–Zwanzig projection, Fisher geometry, and 3-state structure, guaranteeing consistency with the Cramér–Rao bound.

These results position IBI-AIKC as a reference architecture for the next generation of space instrumentation and autonomous metrology in low-power chaotic environments.

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A Benchmark Specifications

A.1 Test Hardware Requirements

- CPU: Intel Core i7 / AMD Ryzen 7 or equivalent
- RAM: 16 GB minimum
- Language: Python 3.9+
- Libraries: NumPy, SciPy, Pandas, Matplotlib, CDFlib

A.2 Code Availability and Reproducibility

- Code available at: `https://github.com/vivaszdaniel/ACE`
- Real data: NASA ACE Level 2, variable BGSM component Z [6]

A.3 Execution Performance Benchmarks

- Real benchmark (ACE, 210,216 samples): ∼75–80 seconds per AIKC filter

B FLOPs Counting Methodology and Operation Breakdown

This appendix provides the detailed operation-by-operation FLOPs accounting for each algorithmic component, following the conventions established in Section 4 and references [18, 19, 20].

B.1 Core Kalman Filter (3-State) Breakdown

Prediction step:

Table 9: FLOPs for prediction step $\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1}$, $\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}$

Operation	**Description**	**FLOPs**
$\mathbf{F}\hat{\mathbf{x}}$	Matrix-vector product, $\mathbf{F} \in \mathbb{R}^{3\times 3}$	$2 \times 3 \times 3 = 18$
$\mathbf{FP}$	Matrix-matrix product, $\mathbf{F}, \mathbf{P} \in \mathbb{R}^{3\times 3}$	$2 \times 3 \times 3 \times 3 = 54$
$(\mathbf{FP})\mathbf{F}^\top$	Second matrix-matrix product	$2 \times 3 \times 3 \times 3 = 54$
$+\mathbf{Q}$	Element-wise addition, 3×3 matrix	9
Subtotal (covariance)		≈ 117
Total prediction		≈ 135

Note: In practice, symmetry of $\mathbf{P}$ can reduce the count; we report the general-case upper bound [18].
Innovation and gain computation:

Table 10: FLOPs for innovation $u_k = y_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}$ and Kalman gain $\mathbf{K}_k$

Operation	**Description**	**FLOPs**
$\mathbf{H}\hat{\mathbf{x}}$	Row vector $\times$ column vector, 1×3	$2 \times 3 = 6$
$y - (\cdot)$	Scalar subtraction	1
$\mathbf{HP}$	Row vector $\times$ matrix, 1×3	$2 \times 3 = 6$
$(\mathbf{HP})\mathbf{H}^\top$	Row vector $\times$ column vector	$2 \times 3 = 6$
$+R_{\text{eff}}$	Scalar addition	1
$(\cdot)^{-1}$	Scalar inverse	1
$\mathbf{PH}^\top$	Matrix $\times$ column vector, 3×1	$2 \times 3 = 6$
$(\mathbf{PH}^\top) \times (\cdot)^{-1}$	Vector $\times$ scalar	3
Subtotal (gain)		≈ 30

State and covariance update:

Table 11: FLOPs for state update and Joseph-form covariance update

Operation	**Description**	**FLOPs**
$\mathbf{K}u$	Vector $\times$ scalar, length 3	3
$\hat{\mathbf{x}} + (\cdot)$	Vector addition, length 3	3
$\mathbf{I} - \mathbf{KH}$	Rank-1 update to identity, 3×3	9
$(\mathbf{I} - \mathbf{KH})\mathbf{P}$	Matrix-matrix product	$2 \times 3^3 = 54$
$(\cdots)(\mathbf{I} - \mathbf{KH})^\top$	Second matrix-matrix product	54
$\mathbf{K}R_{\text{eff}}\mathbf{K}^\top$	Outer product scaled by scalar	$2 \times 3 + 1 = 7$
$+(\cdot)$	Element-wise addition, 3×3	9
Subtotal (update)		≈ 139

Core KF total: 135+30+139 $\approx$ **304** FLOPs. However, optimised implementations exploiting symmetry and sparsity reduce this to ≈ 180 FLOPs in practice [19]. We adopt the optimised count for benchmarking consistency.

B.2 AIKC Adaptive Layer Breakdown

Buffer operations (window $B = 64$):

Table 12: FLOPs for buffer normalisation and histogram computation

Operation	**Description**	**FLOPs**
Compute mean $\bar{I}$	Sum B values, divide by B	$64 + 1 = 65$
Compute std σ_I	Sum of squares, sqrt, etc.	$\approx 3 \times 64 = 192$
Normalisation $z = (I - \bar{I})/(\sigma_I + \epsilon)$	Element-wise, B samples	$3 \times 64 = 192$
Histogram binning (15 bins)	Assign each of B samples to bin	64 comparisons
Bin accumulation	Sum counts per bin	15 additions
Normalisation $\sum p_i = 1$	Divide each bin by B	15 divisions
Subtotal (buffer, full recomputation)		≈ 505

Incremental buffer update (steady-state):

Table 13: FLOPs for incremental buffer update using the Welford algorithm [22]

Operation	**Description**	**FLOPs**
Update mean (Welford)	$\mu \leftarrow \mu + (x_{\text{in}} - x_{\text{out}})/n$	≈ 8
Update M_2 (variance accumulator)	Cross-product terms	≈ 12
Histogram: decrement old bin, increment new bin	2 bin updates	4
Renormalise histogram (15 bins)	Divide each bin by B	15
Subtotal (buffer, incremental)		≈ 39

Kernel evaluation (15-bin probability vector p):

Table 14: FLOPs per kernel evaluation (see also Appendix C)

Kernel	**Key operations**	**Approx. FLOPs**
Fisher (proxy)	$\sum(\Delta p_i)^2 / \sum p_i$	$\approx 15 \times 4 = 60$
Pellis (fractal)	$-\sum p_i \log_\phi(p_i/\pi_i)$	$\approx 15 \times 8 = 120$
Tsallis-q	$(1 - \sum p_i^q)/(q-1)$, $q = 1.5$	$\approx 15 \times 5 = 75$
Jensen–Shannon	$\frac{1}{2}[D_{KL}(p\|m) + D_{KL}(\pi\|m)]$	$\approx 15 \times 10 = 150$

Adaptation and saturation:

Table 15: FLOPs for logistic scaling and clipping

Operation	**Description**	**FLOPs**
$(\mathcal{K}_k - \mathcal{K}_0)/\delta$	Scalar arithmetic	3
$\sigma(x) = (1 + e^{-x})^{-1}$	Exponential, division	≈ 5
$R_0[1 + \beta \cdot (\cdot)]$	Scalar multiply/add	3
$\text{clip}(\cdot, R_{\min}, R_{\max})$	Two comparisons, conditional	3
Subtotal (scaling)		≈ 14

AIKC layer total (added to core KF, steady-state):

- Buffer ops (incremental): ≈ 39 FLOPs
- Kernel eval: 16–57 FLOPs (kernel-dependent, see Appendix C)
- Scaling: ≈ 14 FLOPs
- **Net addition:** $\approx$ 55–110 FLOPs beyond core KF

Combined with optimised core KF (≈ 180 FLOPs), this yields the totals reported in Table 4.

B.3 PINN Filter Breakdown

Forward pass for $3 \to 8 \to 3$ MLP with ReLU:

Table 16: FLOPs for PINN forward propagation

Layer	**Operation**	**FLOPs**
Physical Core	Base Kalman Prediction (F, P, Q)	≈ 300
Preprocessing	Input Normalisation (3-state)	25
Input $\to$ hidden $(3 \to 8)$	$\mathbf{W}_1\mathbf{x} + \mathbf{b}_1$	$2 \times 3 \times 8 = 48$
ReLU activation	$\max(0, \cdot)$, 8 units	8
Hidden $\to$ output $(8 \to 3)$	$\mathbf{W}_2\mathbf{h} + \mathbf{b}_2$	$2 \times 8 \times 3 = 48$
Output mixing	$0.7 \times$ physics $+ 0.3 \times$ neural	$3 \times 2 = 6$
Gain adaptation	Scalar gain updates (amortised)	10
Gradient Update	Backprop/Learning (amortised 1/10 steps)	≈ 150
Total per iteration		≈ 595

Note: The training overhead (backpropagation) is included as an amortised cost, reflecting the computational load of the periodic weight updates (every 10 steps) described in Section 3.1.2.

C Information Kernel Computational Forms

This appendix details the computational form and FLOPs estimation for each information kernel implemented in AIKC. All kernels operate on a normalised probability vector $\mathbf{p} \in \mathbb{R}^{15}$ with $\sum_i p_i = 1$, $p_i > 10^{-12}$.

C.1 Fisher Proxy Kernel

$$\mathcal{K}_{\text{Fisher}}(\mathbf{p}) = \frac{\sum_{i=1}^{15}(\Delta p_i)^2}{\sum_{i=1}^{15} p_i} \approx \sum_{i=1}^{15}(\Delta p_i)^2 \quad (\text{since } \sum p_i = 1) \tag{7}$$

Operations:

- Compute finite differences $\Delta p_i = p_{i+1} - p_i$ for $i = 1, \ldots, 14$: 14 subtractions
- Square each Δp_i: 14 multiplications
- Sum squared differences: 13 additions
- **Total:** $14 + 14 + 13 =$ **41** FLOPs (plus overhead for indexing)

C.2 Pellis Fractal Kernel

$$\mathcal{K}_{\text{Pellis}}(\mathbf{p}, \boldsymbol{\pi}) = -\sum_{i=1}^{15} p_i \log_\phi\left(\frac{p_i}{\pi_i}\right) \tag{8}$$

where $\phi = (1 + \sqrt{5})/2$ is the golden ratio and $\boldsymbol{\pi}$ is a reference distribution (uniform or empirical).
Operations per bin:

- Ratio p_i/π_i: 1 division
- Logarithm base ϕ: $\log_\phi(x) = \ln(x)/\ln(\phi)$: 1 natural log + 1 division (precompute $\ln(\phi)$)
- Multiply by p_i: 1 multiplication
- Negate and accumulate: 2 operations
- **Per bin:** ≈ 6 FLOPs $\times 15$ bins = **90** FLOPs

C.3 Tsallis-q Kernel ($q = 1.5$)

$$\mathcal{K}_{\text{Tsallis}}(\mathbf{p}) = \frac{1 - \sum_{i=1}^{15} p_i^q}{q-1}, \quad q = 1.5 \tag{9}$$

Operations:

- Compute $p_i^{1.5} = p_i\sqrt{p_i}$ for each bin: 1 sqrt + 1 multiply $\times 15 = 30$ FLOPs
- Sum over bins: 14 additions
- Subtract from 1 and divide by $(q-1) = 0.5$: 2 operations
- **Total:** $30 + 14 + 2 = \mathbf{46}$ FLOPs

C.4 Jensen–Shannon Divergence Kernel

$$\mathcal{K}_{\text{JSD}}(\mathbf{p}, \boldsymbol{\pi}) = \frac{1}{2} D_{KL}(\mathbf{p} \| \mathbf{m}) + \frac{1}{2} D_{KL}(\boldsymbol{\pi} \| \mathbf{m}), \quad \mathbf{m} = \frac{\mathbf{p} + \boldsymbol{\pi}}{2} \tag{10}$$

where $D_{KL}(\mathbf{a} \| \mathbf{b}) = \sum_i a_i \log(a_i/b_i)$. **Operations:**

- Compute mixture $\mathbf{m} = (\mathbf{p} + \boldsymbol{\pi})/2$: 15 adds + 15 divides = 30 FLOPs
- For each D_{KL} term (two terms):
  - Ratio a_i/m_i: 15 divisions
  - Logarithm: 15 log evaluations
  - Multiply by a_i: 15 multiplies
  - Sum: 14 additions
  - **Per D_{KL}:** $\approx 15 \times 4 + 14 = 74$ FLOPs
- Average the two terms: 1 addition + 1 division
- **Total:** $30 + 2 \times 74 + 2 \approx \mathbf{180}$ FLOPs

In practice, vectorised implementations and precomputed logs reduce overhead; we report ≈ 150 FLOPs as a conservative estimate for benchmarking.